

Hospital Appointment & Patient Care Management System

Problem Statement & Architectural Approach

DBMS Project Milestone — Review 1

Database Platform: PostgreSQL 16 on Neon Cloud

The Problem: Structural Breakdowns in Healthcare Operations

- • Schedule Collisions & Doctor Double-Booking:
 - - Front desks frequently book overlapping patients into the same consultation slot, leading to long waiting queues and doctor burnout.
- • Fragmented & Inaccessible Patient Records:
 - - Clinical history, past diagnoses, lab reports, and medications are stored across disconnected paper slips, preventing a unified longitudinal chart at point-of-care.
- • Inpatient Bed Allocation Bottlenecks:
 - - Lack of real-time bed state tracking results in double-allocation of ward beds, admission delays, and unrecorded vacancies.
- • Uncoordinated Clinical Orders & Revenue Leakage:
 - - Lab tests and prescriptions are unlinked from billing ledgers, causing untracked dispensing and severe financial leakage.

Why Existing Systems Fail: Root Cause Analysis

- • Absence of Database-Level Concurrency & Integrity Enforcement:
 - - Legacy systems rely on manual checks or fragile client-side scripts rather than database ACID constraints.
- • Unnormalized Data Architecture (Anomalies & Redundancy):
 - - Multi-valued fields (e.g., comma-separated drug lists) violate 1NF, while transitive dependencies create update and deletion anomalies.
- • Disconnected Departmental Silos:
 - - Front-desk scheduling, clinical consultations, pathology labs, and billing operate on isolated tables or paper records.
- • Vulnerability to Invalid Data:
 - - Negative invoices, overlapping bed occupancies, and invalid physiological bounds are accepted without engine-level rejection.

Our Approach: Unified Relational Architecture (PostgreSQL & 3NF)

- • Centralized Relational Schema (16 Normalized Entities in 6 Clusters):
 - - Provider Roster | Patient Appointments | Clinical Care & EMR | Diagnostics | Inpatient Wards | Financials & Billing
- • Strict Third Normal Form (3NF) Normalization:
 - - 1NF: Atomic line items (prescription_item) eliminate multi-valued repeating groups.
 - - 2NF & 3NF: Single surrogate keys and isolated lookup tables eliminate partial and transitive functional dependencies.
- • Engine-Level Defensive Constraints (Active Database Protection):
 - - UNIQUE(doctor_id, date, time) mathematically prevents appointment double-booking.
 - - Real-time Bed State Machine (AVAILABLE, OCCUPIED, MAINTENANCE) prevents double-occupancy.
 - - Domain CHECK bounds enforce positive balances, chronological stay validity, and valid patient vitals.

Strategic Outcomes: Measurable Operational Impact

- • Zero-Collision Scheduling:
 - - 100% elimination of double-booked consultation slots via engine-enforced uniqueness.
- • Point-of-Care Longitudinal EMR:
 - - Clinicians access unified consultation notes, ICD-10 diagnoses, prescribed drugs, and lab results via indexed multi-table joins.
- • Real-Time Inpatient Bed Visibility:
 - - Instant bed occupancy percentages across ICU, Semi-Private, and General wards with exclusive occupancy guarantees.
- • Closed-Loop Financial Reconciliation:
 - - Itemized bills aggregate consultation, lab investigations, and bed per-diem charges into a single ledger with complete multi-tender audit trails.
- • Review 1 Foundation:
 - - Complete conceptual ERD, production PostgreSQL DDL, and realistic seed data ready for Review 2 (Triggers & Views).

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PROBLEM IDENTIFICATION & JUSTIFICATION

PROJECT FOUNDATION

STAKEHOLDER ARCHITECTURE

SYSTEM SPECIFICATION

CONCEPTUAL ARCHITECTURE

DATABASE CONCEPTUAL MODEL

RELATIONAL MODEL

DATABASE THEORY & RIGOR

DATABASE ENGINE PROTECTION

DATA RETRIEVAL & VERIFICATION

DATABASE MANAGEMENT SYSTEMS • REVIEW 1 DELIVERABLE

Relational Schema, Crow's Foot ERD & 3NF Table Architecture

Hospital Appointment and Patient Care Management System (HAPCMS) • PostgreSQL 16 on Neon Cloud with Prisma ORM

RELATIONAL RIGOR

16 Normalized 3NF Entities

Completely decomposed into 6 functional clusters. Zero repeating groups, zero partial dependencies, and zero transitive dependencies.

INTEGRITY & CONCURRENCY

10 Controlled ENUMs & CHECKs

Declarative engine-level double-booking prevention, positive balance rules, chronological bounds, and controlled diagnostic states.

MODERN DATA LAYER

Prisma Studio & Type-Safe ORM

Introspected from PostgreSQL with full foreign key traversal, live GUI on port 5555, and automated vector ER diagram compilation.

STUDENT & COURSE DETAILS

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Course: Relational Database Management Systems (DBMS) • Review 1 (Week 7 — 5 Marks)

PostgreSQL 16.4

Prisma v6.19

100% 3NF

Course: Relational Database Systems

Milestone: Review 2 (Week 11)

5 Marks

FOUNDATION BUILT IN REVIEW 1

REVIEW 2 · NORMALIZED DATABASE DESIGN

FORMAL DEPENDENCY ANALYSIS

REVIEW 2 · DATA DICTIONARY

REVIEW 2 · DDL IMPLEMENTATION

INTEGRITY ENFORCEMENT

REVIEW 2 · SAMPLE DATA

REVIEW 2 · SQL QUERIES

REVIEW 2 · VIEWS & REPORTS

PERFORMANCE OPTIMIZATION

REVIEW 2 · PROTOTYPE STATUS